Fourier Analysis

Problem

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Fourier Analysis: An Introduction

Chapter 1. The Genesis of Fourier Analysis

1. Consider the Dirichlet problem illustrated in Figure 11.

More precisely, we look for a solution of the steady-state heat equation $\triangle u = 0$ in the rectangle $R = \{(x, y) : \ 0 \le x \le \pi, \ \ 0 \le y \le 1\}$ that vanishes on the vertical sides of R, and so that

$$u(x, 0) = f_0(x) \qquad \text{and} \qquad u(x, 1) = f_1(x)\,,$$

where f_0 and f_1 are initial data which fix the temperature distribution on the horizontal sides of the rectangle.

Use separation of variables to show that if f_0 and f_1 have Fourier expansions

$$f_0(x) = \sum_{k=1}^{\infty} A_k \sin kx \qquad \text{and} \qquad f_1(x) = \sum_{k=1}^{\infty} B_k \sin kx,$$

then

$$u(x, y) = \sum_{k=1}^{\infty} \left(\frac{\sinh k(1-y)}{\sinh k} A_k + \frac{\sinh ky}{\sinh k} B_k \right) \sin kx.$$

We recall the definitions of the hyperbolic sine and cosine functions:

$$\sinh x = \frac{e^x - e^{-x}}{2} \qquad \text{and} \qquad \cosh x = \frac{e^x + e^{-x}}{2}.$$

Compare this result with the solution of the Dirichlet problem in the strip obtained in Problem 3, Chapter 5.

Chapter 2. Basic Properties of Fourier Series

1. One can construct Riemann integrable functions on $[0,1]$ that have a dense set of discontinuities as follows.

(a) Let $f(x) = 0$ when $x < 0$, and $f(x) = 1$ if $x \geq 0$. Choose a countable dense sequence $\{r_n\}$ in $[0,1]$. Then, show that the function

$$F(x) = \sum_{n=1}^{\infty} \frac{1}{n^2} f(x - r_n)$$

is integrable and has discontinuities at all points of the sequence $\{r_n\}$. [Hint: F is monotonic and bounded.]

(b) Consider next

$$F(x) = \sum_{n=1}^{\infty} 3^{-n} g(x - r_n),$$

where $g(x) = \sin 1/x$ when $x \neq 0$, and $g(0) = 0$. Then F is integrable, discontinuous at each $x = r_n$, and fails to be monotonic in any subinterval of $[0,1]$. [Hint: Use the fact that $3^{-k} > \sum_{n>k} 3^{-n}$.]

(c) The original example of Riemann is the function

$$F(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2},$$

where $(x) = x$ for $x \in (-1/2, 1/2]$ and (x) is continued to $\mathbb{R}$ by periodicity, that is, $(x+1) = (x)$. It can be shown that F is discontinuous whenever $x = m/2n$, where $m, n \in \mathbb{Z}$ with m odd and $n \neq 0$.

2. Let D_N denote the Dirichlet kernel

$$D_N(\theta) = \sum_{k=-N}^{N} e^{ik\theta} = \frac{\sin((N+1/2)\theta)}{\sin(\theta/2)},$$

and define

$$L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(\theta)|\, d\theta.$$

(a) Prove that

$$L_N \geq c \log N$$

for some constant $c > 0$. [Hint: Show that $|D_N(\theta)| \geq c\frac{\sin((N+1/2)\theta)}{|\theta|}$, change variables, and prove that

$$L_N \geq c \int_{\pi}^{N\pi} \frac{|\sin\theta|}{|\theta|}\, d\theta + O(1).$$

Write the integral as a sum $\sum_{k=1}^{N-1} \int_{k\pi}^{(k+1)\pi}$. To conclude, use the fact that $\sum_{k=1}^{n} 1/k \geq c \log n$.] A more careful estimate gives

$$L_N = \frac{4}{\pi^2} \log N + O(1).$$

(b) Prove the following as a consequence: for each $n \geq 1$, there exists a continuous function f_n such that $|f_n| \leq 1$ and $|S_n(f_n)(0)| \geq c' \log n$. [Hint: The function g_n which is equal to 1 when D_n is positive and -1 when D_n is negative has the desired property but is not continuous. Approximate g_n in the integral norm (in the sense of Lemma 3.2) by continuous functions h_k satisfying $|h_k| \leq 1$.]

3.* Littlewood provided a refinement of Tauber's theorem:

(a) If $\sum c_n$ is Abel summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s.

(b) As a consequence, if $\sum c_n$ is Cesàro summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s.

These results may be applied to Fourier series. By Exercise 17, they imply that if f is an integrable function that satisfies $\hat{f}(\nu) = O(1/|\nu|)$, then:

(i) If f is continuous at θ, then

$$S_N(f)(\theta) \to f(\theta) \qquad \text{as } N \to \infty.$$

(ii) If f has a jump discontinuity at θ, then

$$S_N(f)(\theta) \to \frac{f(\theta^+) + f(\theta^-)}{2} \qquad \text{as } N \to \infty.$$

(iii) If f is continuous on $[-\pi, \pi]$, then $S_N(f) \to f$ uniformly.

For the simpler assertion (b), hence a proof of (i), (ii), and (iii), see Problem 5 in Chapter 4.

Chapter 3. Convergence of Fourier Series

1. For each $0 < \alpha < 1$ the series

$$\sum_{n=1}^{\infty} \frac{\sin nx}{n^{\alpha}}$$

converges for every x but is not the Fourier series of a Riemann integrable function.

(a) If the **conjugate Dirichlet kernel** is defined by

$$\tilde{D}_N(x) = \sum_{|n|\leq N} \operatorname{sign}(x)\, e^{inx} \qquad \text{where } \operatorname{sign}(x) = \begin{cases} 1 & \text{if } n > 0 \\ 0 & \text{if } n = 0 \\ -1 & \text{if } n < 0, \end{cases}$$

then show that

$$\tilde{D}_N(x) = \frac{\cos(x/2) - \cos((N+1/2)x)}{\sin(x/2)},$$

and

$$\int_{-\pi}^{\pi} |\tilde{D}_N(x)|\, dx \leq c \log N.$$

(b) As a result, if f is Riemann integrable, then

$$(f * \tilde{D}_N)(0) = O(\log N).$$

(c) In the present case, this leads to

$$\sum_{n=1}^{N} \frac{1}{n^{\alpha}} = O(\log N),$$

which is a contradiction.

2. An important fact we have proved is that the family $\{e^{inx}\}_{n\in\mathbb{Z}}$ is orthonormal in $\mathcal{R}$ and it is also complete, in the sense that the Fourier series of f converges to f in the norm. In this exercise, we consider another family possessing these same properties.

On $[-1,1]$ define

$$L_n(x) = \frac{d^n}{dx^n}(x^2-1)^n, \qquad n = 0, 1, 2, \ldots.$$

Then L_n is a polynomial of degree n which is called the nth **Legendre polynomial**.

(a) Show that if f is indefinitely differentiable on $[-1,1]$, then

$$\int_{-1}^{1} L_n(x) f(x)\, dx = (-1)^n \int_{-1}^{1} (x^2-1)^n f^{(n)}(x)\, dx.$$

In particular, show that L_n is orthogonal to x^m whenever $m < n$. Hence $\{L_n\}_{n=0}^{\infty}$ is an orthogonal family.

(b) Show that

$$\|L_n\|^2 = \int_{-1}^{1} |L_n(x)|^2\, dx = \frac{(n!)^2 2^{2n+1}}{2n+1}.$$

[Hint: First, note that $\|L_n\|^2 = (-1)^n(2n)!\int_{-1}^{1}(x^2-1)^n\, dx$. Write $(x^2-1)^n = (x-1)^n(x+1)^n$ and integrate by parts n times to calculate this last integral.]

(c) Prove that any polynomial of degree n that is orthogonal to $1, x, x^2, \ldots, x^{n-1}$ is a constant multiple of L_n.

(d) Let $\mathcal{L}_n = L_n/\|L_n\|$, which are the normalized Legendre polynomials. Prove that $\{\mathcal{L}_n\}$ is the family obtained by applying the "Gram-Schmidt process" to $\{1, x, \ldots, x^n, \ldots\}$, and conclude that every Riemann integrable function f on $[-1,1]$ has a **Legendre expansion**

$$\sum_{n=0}^{\infty} \langle f, \mathcal{L}_n\rangle \mathcal{L}_n$$

which converges to f in the mean-square sense.

3. Let α be a complex number not equal to an integer.

(a) Calculate the Fourier series of the 2π-periodic function defined on $[-\pi, \pi]$ by $f(x) = \cos(\alpha x)$.

(b) Prove the following formulas due to Euler:

$$\sum_{n=1}^{\infty} \frac{1}{n^2 - \alpha^2} = \frac{1}{2\alpha^2} - \frac{\pi}{2\alpha \tan(\alpha\pi)}.$$

For all $u \in \mathbb{C} - \pi\mathbb{Z}$,

$$\cot u = \frac{1}{u} + 2\sum_{n=1}^{\infty} \frac{u}{u^2 - n^2\pi^2}.$$

(c) Show that for all $\alpha \in \mathbb{C} - \mathbb{Z}$ we have

$$\frac{\alpha\pi}{\sin(\alpha\pi)} = 1 + 2\alpha^2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2 - \alpha^2}.$$

(d) For all $0 < \alpha < 1$, show that

$$\int_0^{\infty} \frac{t^{\alpha-1}}{t+1}\, dt = \frac{\pi}{\sin(\alpha\pi)}.$$

[Hint: Split the integral as $\int_0^1 + \int_1^{\infty}$ and change variables $t = 1/u$ in the second integral. Now both integrals are of the form

$$\int_0^1 \frac{t^{\gamma-1}}{1+t}\, dt, \qquad 0 < \gamma < 1,$$

which one can show is equal to $\sum_{k=0}^{\infty} \frac{(-1)^k}{k+\gamma}$. Use part (c) to conclude the proof.]

4. In this problem, we find the formula for the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n^k}$$

where k is any even integer. These sums are expressed in terms of the Bernoulli numbers; the related Bernoulli polynomials are discussed in the next problem.

Define the **Bernoulli numbers** B_n by the formula

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n.$$

(a) Show that $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_3 = 0$, $B_4 = -1/30$, and $B_5 = 0$.

(b) Show that for $n \geq 1$ we have

$$B_n = -\frac{1}{n+1} \sum_{k=0}^{n-1} \binom{n+1}{k} B_k.$$

(c) By writing

$$\frac{z}{e^z - 1} = 1 - \frac{z}{2} + \sum_{n=2}^{\infty} \frac{B_n}{n!} z^n,$$

show that $B_n = 0$ if n is odd and > 1. Also prove that

$$z \cot z = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{2^{2n} B_{2n}}{(2n)!} z^{2n}.$$

(d) The **zeta function** is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \text{for all } s > 1.$$

Deduce from the result in (c), and the expression for the cotangent function obtained in the previous problem, that

$$x \cot x = 1 - 2 \sum_{m=1}^{\infty} \frac{\zeta(2m)}{\pi^{2m}} x^{2m}.$$

(e) Conclude that

$$2\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{(2m)!} B_{2m}.$$

5. Define the **Bernoulli polynomials** $B_n(x)$ by the formula

$$\frac{ze^{xz}}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n(x)}{n!} z^n.$$

(a) The functions $B_n(x)$ are polynomials in x and

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}.$$

Show that $B_0(x) = 1$, $B_1(x) = x - 1/2$, $B_2(x) = x^2 - x + 1/6$, and $B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$.

(b) If $n \geq 1$, then

$$B_n(x+1) - B_n(x) = nx^{n-1},$$

and if $n \geq 2$, then

$$B_n(0) = B_n(1) = B_n.$$

(c) Define $S_m(n) = 1^m + 2^m + \cdots + (n-1)^m$. Show that

$$(m+1)S_m(n) = B_{m+1}(n) - B_{m+1}.$$

(d) Prove that the Bernoulli polynomials are the only polynomials that satisfy

(i) $B_0(x) = 1$,

(ii) $B_n'(x) = nB_{n-1}(x)$ for $n \geq 1$,

(iii) $\int_0^1 B_n(x)\,dx = 0$ for $n \geq 1$, and show that from (b) one obtains

$$\int_x^{x+1} B_n(t)\,dt = x^n.$$

(e) Calculate the Fourier series of $B_1(x)$ to conclude that for $0 < x < 1$ we have

$$B_1(x) = x - 1/2 = \frac{-1}{\pi} \sum_{k=1}^{\infty} \frac{\sin(2\pi kx)}{k}.$$

Integrate and conclude that

$$B_{2n}(x) = (-1)^{n+1} \frac{2(2n)!}{(2\pi)^{2n}} \sum_{k=1}^{\infty} \frac{\cos(2\pi kx)}{k^{2n}},$$

$$B_{2n+1}(x) = (-1)^{n+1} \frac{2(2n+1)!}{(2\pi)^{2n+1}} \sum_{k=1}^{\infty} \frac{\sin(2\pi kx)}{k^{2n+1}}.$$

Finally, show that for $0 < x < 1$,

$$B_n(x) = -\frac{n!}{(2\pi i)^n} \sum_{k \neq 0} \frac{e^{2\pi ikx}}{k^n}.$$

We observe that the Bernoulli polynomials are, up to normalization, successive integrals of the sawtooth function.

Chapter 4. Some Applications of Fourier Series

1.* This problem explores another relationship between the geometry of a curve and Fourier series. The diameter of a closed curve Γ parametrized by $\gamma(t) = (x(t), y(t))$ on $[-\pi, \pi]$ is defined by

$$d = \sup_{P,\, Q \in \Gamma} |P - Q| = \sup_{t_1,\, t_2 \in [-\pi, \pi]} |\gamma(t_1) - \gamma(t_2)|.$$

If a_n is the nth Fourier coefficient of $\gamma(t) = x(t) + iy(t)$ and ℓ denotes the length of Γ, then

(a) $2|a_n| \leq d$ for all $n \neq 0$.

(b) $\ell \leq \pi d$, whenever Γ is convex.

Property (a) follows from the fact that $2a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} [\gamma(t) - \gamma(t + \pi/n)] e^{-int}\, dt$.

The equality $\ell = \pi d$ is satisfied when Γ is a circle, but surprisingly, this is not the only case. In fact, one finds that $\ell = \pi d$ is equivalent to $2|a_1| = d$. We re-parametrize γ so that for each t in $[-\pi, \pi]$ the tangent to the curve makes an angle t with the y-axis. Then, if $a_1 = 1$ we have

$$\gamma'(t) = ie^{it}(1 + r(t)),$$

where r is a real-valued function which satisfies $r(t) + r(t + \pi) = 0$, and $|r(t)| \leq 1$. Figure 7 (a) shows the curve obtained by setting $r(t) = \cos 5t$. Also, Figure 7 (b) consists of the curve where $r(t) = h(3t)$, with $h(s) = -1$ if $-\pi \leq s \leq 0$ and $h(s) = 1$ if $0 < s < \pi$. This curve (which is only piecewise of class C^1) is known as the Reuleaux triangle and is the classical example of a convex curve of constant width which is not a circle.

2.* Here we present an estimate of Weyl which leads to some interesting results.

(a) Let $S_N = \sum_{n=1}^{N} e^{2\pi i f(n)}$. Show that for $H \leq N$, one has

$$|S_N|^2 \leq c\,\frac{N}{H}\sum_{h=0}^{H}\left|\sum_{n=1}^{N-h} e^{2\pi i(f(n+h)-f(n))}\right|,$$

for some constant $c > 0$ independent of N, H, and f.

(b) Use this estimate to show that the sequence $\langle n^2\gamma\rangle$ is equidistributed in $[0,1)$ whenever γ is irrational.

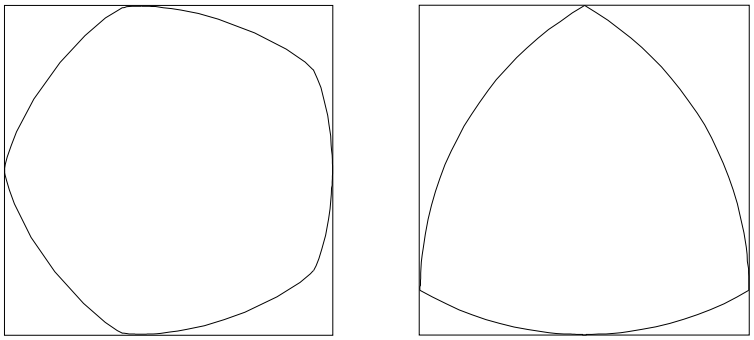


Figure 7. Some curves with maximal length for a given diameter

(c) More generally, show that if $\{\xi_n\}$ is a sequence of real numbers so that for all positive integers h the difference $\langle \xi_{n+h} - \xi_n\rangle$ is equidistributed in $[0,1)$, then $\langle \xi_n\rangle$ is also equidistributed in $[0,1)$.

(d) Suppose that $P(x) = c_n x^n + \cdots + c_0$ is a polynomial with real coefficients, where at least one of $c_1, \ldots, c_n$ is irrational. Then the sequence $\langle P(n)\rangle$ is equidistributed in $[0,1)$.

[Hint: For (a), let $a_n = e^{2\pi i f(n)}$ when $1 \leq n \leq N$ and 0 otherwise. Then write $H\sum_n a_n = \sum_{k=1}^{H}\sum_n a_{n+k}$ and apply the Cauchy-Schwarz inequality. For (b), note that $(n+h)^2\gamma - n^2\gamma = 2nh\gamma + h^2\gamma$, and use the fact that for each integer h, the sequence $\langle 2nh\gamma\rangle$ is equidistributed. Finally, to prove (d), assume first that $P(x) = Q(x) + c_1 x + c_0$ where c_1 is irrational, and estimate the exponential sum $\sum_{n=1}^{N} e^{2\pi i k P(n)}$. Then, argue by induction on the highest degree term which has an irrational coefficient, and use part (c).]

3.* If $\sigma > 0$ is not an integer and $a \neq 0$, then $\langle an^\sigma \rangle$ is equidistributed in $[0, 1)$.
See also Exercise 8.

4. An elementary construction of a continuous but nowhere differentiable function is obtained by "piling up singularities," as follows.

On $[-1, 1]$ consider the function

$$\varphi(x) = |x|$$

and extend φ to $\mathbb{R}$ by requiring it to be periodic of period 2. Clearly, φ is continuous on $\mathbb{R}$ and $|\varphi(x)| \leq 1$ for all x so the function f defined by

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$$

is continuous on $\mathbb{R}$.

(a) Fix $x_0 \in \mathbb{R}$. For every positive integer m, let $\delta_m = \pm\frac{1}{2}4^{-m}$ where the sign is chosen so that no integer lies in between $4^m x_0$ and $4^m(x_0 + \delta_m)$. Consider the quotient

$$\gamma_n = \frac{\varphi(4^n(x_0 + \delta_m)) - \varphi(4^n x_0)}{\delta_m}.$$

Prove that if $n > m$, then $\gamma_n = 0$, and for $0 \leq n \leq m$ one has $|\gamma_n| \leq 4^n$ with $|\gamma_m| = 4^m$.

(b) From the above observations prove the estimate

$$\left|\frac{f(x_0 + \delta_m) - f(x_0)}{\delta_m}\right| \geq \frac{1}{2}(3^m + 1),$$

and conclude that f is not differentiable at x_0.

5. Let f be a Riemann integrable function on the interval $[-\pi, \pi]$. We define the generalized delayed means of the Fourier series of f by

$$\sigma_{N,K} = \frac{S_N + \cdots + S_{N+K-1}}{K}.$$

Note that in particular

$$\sigma_{0,N} = \sigma_N, \qquad \sigma_{N,1} = S_N \qquad \text{and} \qquad \sigma_{N,N} = \Delta_N,$$

where Δ_N are the specific delayed means used in Section 3.

(a) Show that

$$\sigma_{N,K} = \frac{1}{K}\left((N+K)\sigma_{N+K} - N\sigma_N\right),$$

and

$$\sigma_{N,K} = S_N + \sum_{N+1 \le |\nu| \le N+K-1} \left(1 - \frac{|\nu| - N}{K}\right) \hat{f}(\nu) e^{i\nu\theta}.$$

From this last expression for $\sigma_{N,K}$ conclude that

$$|\sigma_{N,K} - S_M| \le \sum_{N+1 \le |\nu| \le N+K-1} |\hat{f}(\nu)|$$

for all $N \le M < N + K$.

(b) Use one of the above formulas and Fejér's theorem to show that with $N = kn$ and $K = n$, then

$$\sigma_{kn,n}(f)(\theta) \to f(\theta) \qquad \text{as } n \to \infty$$

whenever f is continuous at θ, and also

$$\sigma_{kn,n}(f)(\theta) \to \frac{f(\theta^+) + f(\theta^-)}{2} \qquad \text{as } n \to \infty$$

at a jump discontinuity (refer to the preceding chapters and their exercises for the appropriate definitions and results). In the case when f is continuous on $[-\pi, \pi]$, show that $\sigma_{kn,n}(f) \to f$ uniformly as $n \to \infty$.

(c) Using part (a), show that if $\hat{f}(\nu) = O(1/|\nu|)$ and $kn \le m < (k+1)n$, we get

$$|\sigma_{kn,n} - S_m| \le \frac{C}{k} \qquad \text{for some constant } C > 0.$$

(d) Suppose that $\hat{f}(\nu) = O(1/|\nu|)$. Prove that if f is continuous at θ then

$$S_N(f)(\theta) \to f(\theta) \qquad \text{as } N \to \infty,$$

and if f has a jump discontinuity at θ then

$$S_N(f)(\theta) \to \frac{f(\theta^+) + f(\theta^-)}{2} \qquad \text{as } N \to \infty.$$

Also, show that if f is continuous on $[-\pi, \pi]$, then $S_N(f) \to f$ uniformly.

(e) The above arguments show that if $\sum c_n$ is Cesàro summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s. This is a weak version of Littlewood's theorem (Problem 3, Chapter 2).

6. Dirichlet's theorem states that the Fourier series of a real continuous periodic function f which has only a finite number of relative maxima and minima converges everywhere to f (and uniformly).

Prove this theorem by showing that such a function satisfies $\hat{f}(n) = O(1/|n|)$. [Hint: Argue as in Exercise 17, Chapter 3; then use conclusion (d) in Problem 5 above.]

Chapter 5. The Fourier Transform on R

1. The equation

$$x^2 \frac{\partial^2 u}{\partial x^2} + ax \frac{\partial u}{\partial x} = \frac{\partial u}{\partial t} \tag{17}$$

with $u(x,0) = f(x)$ for $0 < x < \infty$ and $t > 0$ is a variant of the heat equation which occurs in a number of applications. To solve (17), make the change of variables $x = e^{-y}$ so that $-\infty < y < \infty$. Set $U(y,t) = u(e^{-y},t)$ and $F(y) = f(e^{-y})$. Then the problem reduces to the equation

$$\frac{\partial^2 U}{\partial y^2} + (1-a)\frac{\partial U}{\partial y} = \frac{\partial U}{\partial t},$$

with $U(y,0) = F(y)$. This can be solved like the usual heat equation (the case $a = 1$) by taking the Fourier transform in the y variable. One must then compute the integral $\int_{-\infty}^{\infty} e^{(-4\pi^2\xi^2 + (1-a)2\pi i\xi)t} e^{2\pi i \xi v}\, d\xi$. Show that the solution of the original problem is then given by

$$u(x,t) = \frac{1}{(4\pi t)^{1/2}} \int_0^{\infty} e^{-(\log(v/x)+(1-a)t)^2/(4t)} f(v)\, \frac{dv}{v}.$$

2. The **Black-Scholes** equation from finance theory is

$$\frac{\partial V}{\partial t} + rs\frac{\partial V}{\partial s} + \frac{\sigma^2 s^2}{2}\frac{\partial^2 V}{\partial s^2} - rV = 0, \qquad 0 < t < T, \tag{18}$$

subject to the "final" boundary condition $V(s,T) = F(s)$. An appropriate change of variables reduces this to the equation in Problem 1. Alternatively, the substitution $V(s,t) = e^{ax+b\tau}U(x,\tau)$ where $x = \log s$, $\tau = \frac{\sigma^2}{2}(T-t)$, $a = \frac{1}{2} - \frac{r}{\sigma^2}$, and $b = -\left(\frac{1}{2} + \frac{r}{\sigma^2}\right)^2$ reduces (18) to the one-dimensional heat equation with the initial condition $U(x,0) = e^{-ax}F(e^x)$. Thus a solution to the Black-Scholes equation is

$$V(s,t) = \frac{e^{-r(T-t)}}{\sqrt{2\pi\sigma^2(T-t)}} \int_0^\infty e^{-\frac{(\log(s/s^*)+(r-\sigma^2/2)(T-t))^2}{2\sigma^2(T-t)}} F(s^*)\, ds^*.$$

3. * **The Dirichlet problem in a strip**. Consider the equation $\triangle u = 0$ in the horizontal strip

$$\{(x, y) : 0 < y < 1, \ -\infty < x < \infty\}$$

with boundary conditions $u(x, 0) = f_0(x)$ and $u(x, 1) = f_1(x)$, where f_0 and f_1 are both in the Schwartz space.

(a) Show (formally) that if u is a solution to this problem, then

$$\hat{u}(\xi, y) = A(\xi)e^{2\pi\xi y} + B(\xi)e^{-2\pi\xi y}.$$

Express A and B in terms of $\widehat{f_0}$ and $\widehat{f_1}$, and show that

$$\hat{u}(\xi, y) = \frac{\sinh(2\pi(1-y)\xi)}{\sinh(2\pi\xi)}\widehat{f_0}(\xi) + \frac{\sinh(2\pi y\xi)}{\sinh(2\pi\xi)}\widehat{f_0}(\xi).$$

(b) Prove as a result that

$$\int_{-\infty}^{\infty} |u(x, y) - f_0(x)|^2 \, dx \to 0 \qquad \text{as } y \to 0$$

and

$$\int_{-\infty}^{\infty} |u(x, y) - f_1(x)|^2 \, dx \to 0 \qquad \text{as } y \to 1.$$

(c) If $\Phi(\xi) = (\sinh 2\pi a\xi)/(\sinh 2\pi\xi)$, with $0 \le a < 1$, then Φ is the Fourier transform of φ where

$$\varphi(x) = \frac{\sin \pi a}{2} \cdot \frac{1}{\cosh \pi x + \cos \pi a}.$$

This can be shown, for instance, by using contour integration and the residue formula from complex analysis (see Book II, Chapter 3).

(d) Use this result to express u in terms of Poisson-like integrals involving f_0 and f_1 as follows:

$$u(x, y) = \frac{\sin \pi y}{2}\left(\int_{-\infty}^{\infty} \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y} \, dt + \int_{-\infty}^{\infty} \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y} \, dt\right).$$

(e) Finally, one can check that the function $u(x, y)$ defined by the above expression is harmonic in the strip, and converges uniformly to $f_0(x)$ as $y \to 0$, and to $f_1(x)$ as $y \to 1$. Moreover, one sees that $u(x, y)$ vanishes at infinity, that is, $\lim_{|x|\to\infty} u(x, y) = 0$, uniformly in y.

In Exercise 12, we gave an example of a function that satisfies the heat equation in the upper half-plane, with boundary value 0, but which was not identically 0. We observed in this case that u was in fact not continuous up to the boundary.

In Problem 4 we exhibit examples illustrating non-uniqueness, but this time with continuity up to the boundary $t = 0$. These examples satisfy a growth condition at infinity, namely $|u(x, t)| \le Ce^{cx^{2+\epsilon}}$, for any $\epsilon > 0$. Problems 5 and 6 show that under the more restrictive growth condition $|u(x, t)| \le Ce^{cx^2}$, uniqueness does hold.

4.* If g is a smooth function on $\mathbb{R}$, define the formal power series

$$u(x,t) = \sum_{n=0}^{\infty} g^{(n)}(t) \frac{x^{2n}}{(2n!)}. \tag{19}$$

(a) Check formally that u solves the heat equation.

(b) For $a > 0$, consider the function defined by

$$g(t) = \begin{cases} e^{-t^{-a}} & \text{if } t > 0 \\ 0 & \text{if } t \leq 0. \end{cases}$$

One can show that there exists $0 < \theta < 1$ depending on a so that

$$|g^{(k)}(t)| \leq \frac{k!}{(\theta t)^k} e^{-\frac{1}{2}t^{-a}} \qquad \text{for } t > 0.$$

(c) As a result, for each x and t the series (19) converges; u solves the heat equation; u vanishes for $t = 0$; and u satisfies the estimate $|u(x,t)| \leq Ce^{c|x|^{2a/(a-1)}}$ for some constants $C, c > 0$.

(d) Conclude that for every $\epsilon > 0$ there exists a non-zero solution to the heat equation which is continuous for $x \in \mathbb{R}$ and $t \geq 0$, which satisfies $u(x,0) = 0$ and $|u(x,t)| \leq Ce^{c|x|^{2+\epsilon}}$.

5.* The following "maximum principle" for solutions of the heat equation will be used in the next problem.

Theorem. *Suppose that $u(x,t)$ is a real-valued solution of the heat equation in the upper half-plane, which is continuous on its closure. Let R denote the rectangle*

$$R = \{(x,y) \in \mathbb{R}^2 : \ a \le x \le b, \ 0 \le t \le c\}$$

and $\partial' R$ be the part of the boundary of R which consists of the two vertical sides and its base on the line $t = 0$ (see Figure 3*). Then*

$$\min_{(x,t)\in\partial' R} u(x,t) = \min_{(x,t)\in R} u(x,t) \qquad \textit{and} \qquad \max_{(x,t)\in\partial' R} u(x,t) = \max_{(x,t)\in R} u(x,t).$$

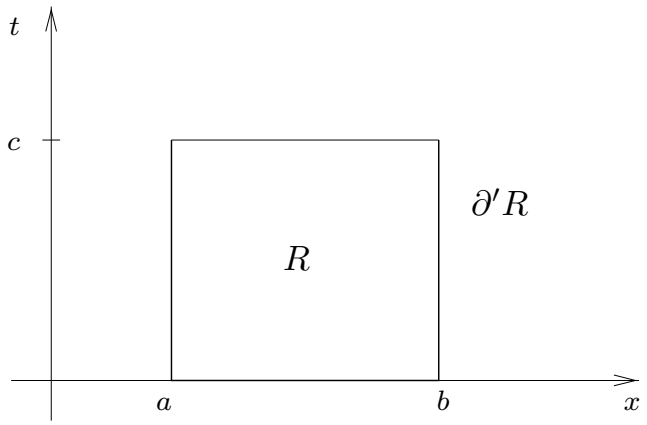


Figure 3. The rectangle R and part of its boundary $\partial' R$

The steps leading to a proof of this result are outlined below.

(a) Show that it suffices to prove that if $u \ge 0$ on $\partial' R$, then $u \ge 0$ in R.

(b) For $\epsilon > 0$, let

$$v(x,t) = u(x,t) + \epsilon t.$$

Then, v has a minimum on R, say at (x_1, t_1). Show that $x_1 = a$ or b, or else $t_1 = 0$. To do so, suppose on the contrary that $a < x_1 < b$ and $0 < t_1 \le c$, and prove that $v_{xx}(x_1,t_1) - v_t(x_1,t_1) \le -\epsilon$. However, show also that the left-hand side must be non-negative.

(c) Deduce from (b) that $u(x,t) \ge \epsilon(t_1 - t)$ for any $(x,t) \in R$ and let $\epsilon \to 0$.

6.* The examples in Problem 4 are optimal in the sense of the following uniqueness theorem due to Tychonoff.

Theorem. *Suppose $u(x,t)$ satisfies the following conditions:*

(i) *$u(x,t)$ solves the heat equation for all $x \in \mathbb{R}$ and and all $t > 0$.*

(ii) *$u(x,t)$ is continuous for all $x \in \mathbb{R}$ and $0 \le t \le c$.*

(iii) *$u(x,0) = 0$.*

(iv) *$|u(x,t)| \le Me^{ax^2}$ for some M, a, and all $x \in \mathbb{R}$, $0 \le t < c$.*

Then u is identically equal to 0.

7.* The **Hermite functions** $h_k(x)$ are defined by the generating identity

$$\sum_{k=0}^{\infty} h_k(x)\frac{t^k}{k!} = e^{-(x^2/2-2tx+t^2)}.$$

(a) Show that an alternate definition of the Hermite functions is given by the formula

$$h_k(x) = (-1)^k e^{x^2/2}\left(\frac{d}{dx}\right)^k e^{-x^2}.$$

[Hint: Write $e^{-(x^2/2-2tx+t^2)} = e^{x^2/2}e^{-(x-t)^2}$ and use Taylor's formula.] Conclude from the above expression that each $h_k(x)$ is of the form $P_k(x)e^{-x^2/2}$, where P_k is a polynomial of degree k. In particular, the Hermite functions belong to the Schwartz space and $h_0(x) = e^{-x^2/2}$, $h_1(x) = 2xe^{-x^2/2}$.

(b) Prove that the family $\{h_k\}_{k=0}^{\infty}$ is complete in the sense that if f is a Schwartz function, and

$$(f, h_k) = \int_{-\infty}^{\infty} f(x)h_k(x)\,dx = 0 \qquad \text{for all } k \geq 0,$$

then $f = 0$. [Hint: Use Exercise 8.]

(c) Define $h_k^*(x) = h_k((2\pi)^{1/2}x)$. Then

$$\widehat{h_k^*}(\xi) = (-i)^k h_k^*(\xi).$$

Therefore, each h_k^* is an eigenfunction for the Fourier transform.

(d) Show that h_k is an eigenfunction for the operator defined in Exercise 23, and in fact, prove that

$$Lh_k = (2k+1)h_k.$$

In particular, we conclude that the functions h_k are mutually orthogonal for the L^2 inner product on the Schwartz space.

(e) Finally, show that $\int_{-\infty}^{\infty}[h_k(x)]^2\,dx = \pi^{1/2}2^k k!$. [Hint: Square the generating relation.]

8.* To refine the results in Chapter 4, and to prove that

$$f_\alpha(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{2\pi i 2^n x}$$

is nowhere differentiable even in the case $\alpha = 1$, we need to consider a variant of the delayed means $\triangle_N$, which in turn will be analyzed by the Poisson summation formula.

(a) Fix an indefinitely differentiable function Φ satisfying

$$\Phi(\xi) = \begin{cases} 1 & \text{when } |\xi| \leq 1, \\ 0 & \text{when } |\xi| \geq 2. \end{cases}$$

By the Fourier inversion formula, there exists $\varphi \in \mathcal{S}$ so that $\hat{\varphi}(\xi) = \Phi(\xi)$. Let $\varphi_N(x) = N\varphi(Nx)$ so that $\widehat{\varphi_N}(\xi) = \Phi(\xi/N)$. Finally, set

$$\tilde{\triangle}_N(x) = \sum_{n=-\infty}^{\infty} \varphi_N(x+n).$$

Observe by the Poisson summation formula that $\tilde{\triangle}_N(x) = \sum_{n=-\infty}^{\infty} \Phi(n/N) e^{2\pi i n x}$, thus $\tilde{\triangle}_N$ is a trigonometric polynomial of degree $\leq 2N$, with terms whose coefficients are 1 when $|n| \leq N$. Let

$$\tilde{\triangle}_N(f) = f * \tilde{\triangle}_N.$$

Note that

$$S_N(f_\alpha) = \tilde{\triangle}_{N'}(f_\alpha)$$

where N' is the largest integer of the form 2^k with $N' \leq N$.

(b) If we set $\tilde{\triangle}_N(x) = \varphi_N(x) + E_N(x)$ where

$$E_N(x) = \sum_{|n| \geq 1} \varphi_N(x+n),$$

then one sees that:

(i) $\sup_{|x| \leq 1/2} |E_N'(x)| \to 0$ as $N \to \infty$.

(ii) $|\tilde{\triangle}_N'(x)| \leq cN^2$.

(iii) $|\tilde{\triangle}_N'(x)| \leq c/(N|x|^3)$, for $|x| \leq 1/2$.

Moreover, $\int_{|x| \leq 1/2} \tilde{\triangle}_N'(x)\, dx = 0$, and $-\int_{|x| \leq 1/2} x\tilde{\triangle}_N'(x)\, dx \to 1$ as $N \to \infty$.

(c) The above estimates imply that if $f'(x_0)$ exists, then

$$(f * \tilde{\triangle}_N')(x_0 + h_N) \to f'(x_0) \qquad \text{as } N \to \infty,$$

whenever $|h_N| \leq C/N$. Then, conclude that both the real and imaginary parts of f_1 are nowhere differentiable, as in the proof given in Section 3, Chapter 4.

Chapter 6. The Fourier Transform on $\mathbf{R}^d$

1. Let J_n denote the nth order Bessel function, for $n \in \mathbb{Z}$. Prove that

(a) $J_n(\rho)$ is real for all real ρ.

(b) $J_{-n}(\rho) = (-1)^n J_n(\rho)$.

(c) $2J_n'(\rho) = J_{n-1}(\rho) - J_{n+1}(\rho)$.

(d) $\left(\frac{2n}{\rho}\right) J_n(\rho) = J_{n-1}(\rho) + J_{n+1}(\rho)$.

(e) $(\rho^{-n} J_n(\rho))' = -\rho^{-n} J_{n+1}(\rho)$.

(f) $(\rho^n J_n(\rho))' = \rho^n J_{n-1}(\rho)$.

(g) $J_n(\rho)$ satisfies the second order differential equation

$$J_n''(\rho) + \rho^{-1} J_n'(\rho) + (1 - n^2/\rho^2) J_n(\rho) = 0.$$

(h) Show that

$$J_n(\rho) = \left(\frac{\rho}{2}\right)^n \sum_{m=0}^{\infty} (-1)^m \frac{\rho^{2m}}{2^{2m} m!(n+m)!}.$$

(i) Show that for all integers n and all real numbers a and b we have

$$J_n(a+b) = \sum_{\ell \in \mathbb{Z}} J_\ell(a) J_{n-\ell}(b).$$

2. Another formula for $J_n(\rho)$ that allows one to define Bessel functions for non-integral values of n, $(n > -1/2)$ is

$$J_n(\rho) = \frac{(\rho/2)^n}{\Gamma(n+1/2)\sqrt{\pi}} \int_{-1}^{1} e^{i\rho t}(1-t^2)^{n-(1/2)}\, dt.$$

(a) Check that the above formula agrees with the definition of $J_n(\rho)$ for integral $n \geq 0$. [Hint: Verify it for $n = 0$ and then check that both sides satisfy the recursion formula (e) in Problem 1.]

(b) Note that $J_{1/2}(\rho) = \sqrt{\frac{2}{\pi}}\rho^{-1/2}\sin\rho$.

(c) Prove that

$$\lim_{n\to -1/2} J_n(\rho) = \sqrt{\frac{2}{\pi}}\rho^{-1/2}\cos\rho.$$

(d) Observe that the formulas we have proved in the text giving F_0 in terms of f_0 (when describing the Fourier transform of a radial function) take the form

$$F_0(\rho) = 2\pi\rho^{-(d/2)+1}\int_0^\infty J_{(d/2)-1}(2\pi\rho r) f_0(r) r^{d/2}\, dr, \tag{17}$$

for $d = 1, 2$, and 3, if one uses the formulas above with the understanding that $J_{-1/2}(\rho) = \lim_{n\to -1/2} J_n(\rho)$. It turns out that the relation between F_0 and f_0 given by (17) is valid in all dimensions d.

3. We observed that the solution $u(x,t)$ of the Cauchy problem for the wave equation given by formula (3) depends only on the initial data on the base on the backward light cone. It is natural to ask if this property is shared by *any* solution of the wave equation. An affirmative answer would imply uniqueness of the solution.

Let $B(x_0, r_0)$ denote the closed ball in the hyperplane $t = 0$ centered at x_0 and of radius r_0. The **backward light cone** with base $B(x_0, r_0)$ is defined by

$$\mathcal{L}_{B(x_0,r_0)} = \{(x,t) \in \mathbb{R}^d \times \mathbb{R} : |x - x_0| \leq r_0 - t, \quad 0 \leq t \leq r_0\}.$$

Theorem *Suppose that $u(x,t)$ is a C^2 function on the closed upper half-plane $\{(x,t) : \ x \in \mathbb{R}^d, t \geq 0\}$ that solves the wave equation*

$$\frac{\partial^2 u}{\partial t^2} = \triangle u.$$

If $u(x,0) = \frac{\partial u}{\partial t}(x,0) = 0$ for all $x \in B(x_0, r_0)$, then $u(x,t) = 0$ for all $(x,t) \in \mathcal{L}_{B(x_0,r_0)}$.

In words, if the initial data of the Cauchy problem for the wave equation vanishes on a ball B, then *any* solution u of the problem vanishes in the backward light cone with base B. The following steps outline a proof of the theorem.

(a) Assume that u is real. For $0 \leq t \leq r_0$ let $B_t(x_0, r_0) = \{x : |x - x_0| \leq r_0 - t\}$, and also define

$$\nabla u(x,t) = \left(\frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_d}, \frac{\partial u}{\partial t} \right).$$

Now consider the energy integral

$$\begin{aligned} E(t) &= \frac{1}{2} \int_{B_t(x_0,r_0)} |\nabla u|^2 \, dx \\ &= \frac{1}{2} \int_{B_t(x_0,r_0)} \left(\frac{\partial u}{\partial t} \right)^2 + \sum_{j=1}^{d} \left(\frac{\partial u}{\partial x_j} \right)^2 \, dx. \end{aligned}$$

Observe that $E(t) \geq 0$ and $E(0) = 0$. Prove that

$$E'(t) = \int_{B_t(x_0,r_0)} \frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial t^2} + \sum_{j=1}^{d} \frac{\partial u}{\partial x_j} \frac{\partial^2 u}{\partial x_j \partial t} \, dx - \frac{1}{2} \int_{\partial B_t(x_0,r_0)} |\nabla u|^2 \, d\sigma(\gamma).$$

(b) Show that

$$\frac{\partial}{\partial x_j} \left[\frac{\partial u}{\partial x_j} \frac{\partial u}{\partial t} \right] = \frac{\partial u}{\partial x_j} \frac{\partial^2 u}{\partial x_j \partial t} + \frac{\partial^2 u}{\partial x_j^2} \frac{\partial u}{\partial t}.$$

(c) Use the last identity, the divergence theorem, and the fact that u solves the wave equation to prove that

$$E'(t) = \int_{\partial B_t(x_0,r_0)} \sum_{j=1}^{d} \frac{\partial u}{\partial x_j} \frac{\partial u}{\partial t} \nu_j \, d\sigma(\gamma) - \frac{1}{2} \int_{\partial B_t(x_0,r_0)} |\nabla u|^2 \, d\sigma(\gamma),$$

where ν_j denotes the jth coordinate of the outward normal to $B_t(x_0, r_0)$.

(d) Use the Cauchy-Schwarz inequality to conclude that

$$\sum_{j=1}^{d} \frac{\partial u}{\partial x_j} \frac{\partial u}{\partial t} \nu_j \leq \frac{1}{2} |\nabla u|^2,$$

and as a result, $E'(t) \leq 0$. Deduce from this that $E(t) = 0$ and $u = 0$.

4.* There exist formulas for the solution of the Cauchy problem for the wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2} \qquad \text{with } u(x,0) = f(x) \text{ and } \frac{\partial u}{\partial t}(x,0) = g(x)$$

in $\mathbb{R}^d \times \mathbb{R}$ in terms of spherical means which generalize the formula given in the text for $d = 3$. In fact, the solution for even dimensions is deduced from that for odd dimensions, so we discuss this case first.

Suppose that $d > 1$ is odd and let $h \in \mathcal{S}(\mathbb{R}^d)$. The spherical mean of h on the ball centered at x of radius t is defined by

$$M_r h(x) = Mh(x,r) = \frac{1}{A_d} \int_{S^{d-1}} h(x - r\gamma)\, d\sigma(\gamma),$$

where A_d denotes the area of the unit sphere S^{d-1} in $\mathbb{R}^d$.

(a) Show that

$$\triangle_x Mh(x,r) = \left[\partial_r^2 + \frac{d-1}{r}\right] Mh(x,r),$$

where $\triangle_x$ denotes the Laplacian in the space variables x, and $\partial_r = \partial/\partial r$.

(b) Show that a twice differentiable function $u(x,t)$ satisfies the wave equation if and only if

$$\left[\partial_r^2 + \frac{d-1}{r}\right] Mu(x,r,t) = \partial_t^2 Mu(x,r,t),$$

where $Mu(x,r,t)$ denote the spherical means of the function $u(x,t)$.

(c) If $d = 2k+1$, define $T\varphi(r) = (r^{-1}\partial_r)^{k-1}[r^{2k-1}\varphi(r)]$, and let $\tilde{u} = TMu$. Then this function solves the one-dimensional wave equation for each fixed x:

$$\partial_t^2 \tilde{u}(x,r,t) = \partial_r^2 \tilde{u}(x,r,t).$$

One can then use d'Alembert's formula to find the solution $\tilde{u}(x,r,t)$ of this problem expressed in terms of the initial data.

(d) Now show that

$$u(x,t) = Mu(x,0,t) = \lim_{r \to 0} \frac{\tilde{u}(x,r,t)}{\alpha r}$$

where $\alpha = 1 \cdot 3 \cdots (d-2)$.

(e) Conclude that the solution of the Cauchy problem for the d-dimensional wave equation, when $d > 1$ is odd, is

$$u(x,t) = \frac{1}{1 \cdot 3 \cdots (d-2)} \left[\partial_t (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} M_t f(x)\right) + \right.$$

$$\left. (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} M_t g(x)\right)\right].$$

5.* The method of descent can be used to prove that the solution of the Cauchy problem for the wave equation in the case when d is even is given by the formula

$$u(x,t) = \frac{1}{1 \cdot 3 \cdots (d-2)} \left[\partial_t (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} \widetilde{M}_t f(x) \right) + \right.$$

$$\left. (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} \widetilde{M}_t g(x) \right) \right],$$

where $\widetilde{M}_t$ denotes the modified spherical means defined by

$$\widetilde{M}_t h(x) = \frac{2}{A_{d+1}} \int_{B^d} \frac{f(x+ty)}{\sqrt{1-|y|^2}}\, dy.$$

6.* Given initial data f and g of the form

$$f(x) = F(x \cdot \gamma) \qquad \text{and} \qquad g(x) = G(x \cdot \gamma),$$

check that the plane wave given by

$$u(x,t) = \frac{F(x \cdot \gamma + t) + F(x \cdot \gamma - t)}{2} + \frac{1}{2} \int_{x \cdot \gamma - t}^{x \cdot \gamma + t} G(s)\, ds$$

is a solution of the Cauchy problem for the d-dimensional wave equation.

In general, the solution is given as a superposition of plane waves. For the case $d = 3$, this can be expressed in terms of the Radon transform as follows. Let

$$\tilde{\mathcal{R}}(f)(t,\gamma) = -\frac{1}{8\pi^2} \left(\frac{d}{dt}\right)^2 R(f)(t,\gamma).$$

Then $u(x,t) =$

$$\frac{1}{2} \int_{S^2} \left[\tilde{\mathcal{R}}(f)(x \cdot \gamma - t, \gamma) + \tilde{\mathcal{R}}(f)(x \cdot \gamma + t, \gamma) + \int_{x \cdot \gamma - t}^{x \cdot \gamma + t} \tilde{\mathcal{R}}(g)(s,\gamma)\, ds \right] d\sigma(\gamma).$$

7. For every real number $a > 0$, define the operator $(-\triangle)^a$ by the formula

$$(-\triangle)^a f(x) = \int_{\mathbb{R}^d} (2\pi|\xi|)^{2a} \hat{f}(\xi) e^{2\pi i \xi \cdot x}\, d\xi$$

whenever $f \in \mathcal{S}(\mathbb{R}^d)$.

(a) Check that $(-\triangle)^a$ agrees with the usual definition of the ath power of $-\triangle$ (that is, a compositions of minus the Laplacian) when a is a positive integer.

(b) Verify that $(-\triangle)^a(f)$ is indefinitely differentiable.

(c) Prove that if a is not an integer, then in general $(-\triangle)^a(f)$ is not rapidly decreasing.

(d) Let $u(x, y)$ be the solution of the steady-state heat equation

$$\frac{\partial^2 u}{\partial y^2} + \sum_{j=1}^{d} \frac{\partial^2 u}{\partial x_j^2} = 0, \qquad \text{with } u(x, 0) = f(x),$$

given by convolving f with the Poisson kernel (see Exercise 8). Check that

$$(-\triangle)^{1/2} f(x) = -\lim_{y \to 0} \frac{\partial u}{\partial y}(x, y),$$

and more generally that

$$(-\triangle)^{k/2} f(x) = (-1)^k \lim_{y \to 0} \frac{\partial^k u}{\partial y^k}(x, y)$$

for any positive integer k.

8.* The reconstruction formula for the Radon transform in $\mathbb{R}^d$ is as follows:

(a) When $d = 2$,

$$\frac{(-\triangle)^{1/2}}{4\pi}\mathcal{R}^*(\mathcal{R}(f)) = f,$$

where $(-\triangle)^{1/2}$ is defined in Problem 7.

(b) If the Radon transform and its dual are defined by analogy to the cases $d = 2$ and $d = 3$, then for general d,

$$\frac{(2\pi)^{1-d}}{2}(-\triangle)^{(d-1)/2}\mathcal{R}^*(\mathcal{R}(f)) = f.$$

Chapter 7. Finite Fourier Analysis

1. Prove that if n and m are two positive integers that are relatively prime, then

$$\mathbb{Z}(nm) \approx \mathbb{Z}(n) \times \mathbb{Z}(m).$$

[Hint: Consider the map $\mathbb{Z}(nm) \to \mathbb{Z}(n) \times \mathbb{Z}(m)$ given by $k \mapsto (k \bmod n, k \bmod m)$, and use the fact that there exist integers x and y such that $xn + ym = 1$.]

2.* Every finite abelian group G is isomorphic to a direct product of cyclic groups. Here are two more precise formulations of this theorem.

- If $p_1, \ldots, p_s$ are the distinct primes appearing in the factorization of the order of G, then

$$G \approx G(p_1) \times \cdots \times G(p_s),$$

  where each $G(p)$ is of the form $G(p) = \mathbb{Z}(p^{r_1}) \times \cdots \times \mathbb{Z}(p^{r_\ell})$, with $0 \leq r_1 \leq \cdots \leq r_\ell$ (this sequence of integers depends on p of course). This decomposition is unique.

- There exist unique integers $d_1, \ldots, d_k$ such that

$$d_1 | d_2, \quad d_2 | d_3, \quad \cdots, \quad d_{k-1} | d_k$$

  and

$$G \approx \mathbb{Z}(d_1) \times \cdots \times \mathbb{Z}(d_k).$$

Deduce the second formulation from the first.

3. Let $\hat{G}$ denote the collection of distinct characters of the finite abelian group G.

(a) Note that if $G = \mathbb{Z}(N)$, then $\hat{G}$ is isomorphic to G.

(b) Prove that $\widehat{G_1 \times G_2} = \hat{G}_1 \times \hat{G}_2$.

(c) Prove using Problem 2 that if G is a finite abelian group, then $\hat{G}$ is isomorphic to G.

4.* When p is prime the group $\mathbb{Z}^*(p)$ is cyclic, and $\mathbb{Z}^*(p) \approx \mathbb{Z}(p-1)$.

Chapter 8. Dirichlet's Theorem

1.* Here are other series that can be summed by the methods in Exercise 15.

(a) For the non-trivial Dirichlet character modulo 6, $L(1,\chi)$ equals

$$\frac{\pi}{2\sqrt{3}} = 1 - \frac{1}{5} + \frac{1}{7} - \frac{1}{11} + \frac{1}{13} + \cdots .$$

(b) If χ is the odd Dirichlet character modulo 8, then $L(1,\chi)$ equals

$$\frac{\pi}{2\sqrt{2}} = 1 + \frac{1}{3} - \frac{1}{5} - \frac{1}{7} + \frac{1}{9} + \frac{1}{11} \cdots .$$

(c) For an odd Dirichlet character modulo 7, $L(1,\chi)$ equals

$$\frac{\pi}{\sqrt{7}} = 1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} - \frac{1}{6} \cdots .$$

(d) For an even Dirichlet character modulo 8, $L(1,\chi)$ equals

$$\frac{\log(1+\sqrt{2})}{\sqrt{2}} = 1 - \frac{1}{3} - \frac{1}{5} + \frac{1}{7} + \frac{1}{9} - \frac{1}{11} \cdots .$$

(e) For an even Dirichlet character modulo 5, $L(1,\chi)$ equals

$$\frac{2}{\sqrt{5}} \log\left(\frac{1+\sqrt{5}}{2}\right) = 1 - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{6} - \frac{1}{7} - \frac{1}{8} + \frac{1}{9} + \frac{1}{11} \cdots .$$

2. Let $d(k)$ denote the number of positive divisors of k.

(a) Show that if $k = p_1^{a_1} \cdots p_n^{a_n}$ is the prime factorization of k, then

$$d(k) = (a_1 + 1) \cdots (a_n + 1).$$

Although Theorem 3.12 shows that on "average" $d(k)$ is of the order of $\log k$, prove the following on the basis of (a):

(b) $d(k) = 2$ for infinitely many k.

(c) For any positive integer N, there is a constant $c > 0$ so that $d(k) \geq c(\log k)^N$ for infinitely many k. [Hint: Let $p_1, \ldots, p_N$ be N distinct primes, and consider k of the form $(p_1 p_2 \cdots p_N)^m$ for $m = 1, 2, \ldots$.]

3. Show that if p is relatively prime to q, then

$$\prod_{\chi} \left(1 - \frac{\chi(p)}{p^s}\right) = \left(\frac{1}{1 - p^{fs}}\right)^g,$$

where $g = \varphi(q)/f$, and f is the order of p in $\mathbb{Z}^*(q)$ (that is, the smallest n for which $p^n \equiv 1 \mod q$). Here the product is taken over all Dirichlet characters modulo q.

4. Prove as a consequence of the previous problem that

$$\prod_{\chi} L(s,\chi) = \sum_{n\geq 1} \frac{a_n}{n^s},$$

where $a_n \geq 0$, and the product is over all Dirichlet characters modulo q.